

CHANGE-POINT ANALYSIS OF HIGH-DIMENSIONAL DATA BASED ON CLUSTERING

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CERTIFICATE

This is to certify that Banashree Ghosh, Master of Technology student in Computer and Communication Sciences Division, has completed bonafied work on the thesis entitled '**Change-point Analysis of High-dimensional Data Based on Clustering**' under my supervision and guidance.

June 2023
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ABSTRACT

Change-point analysis involves examining a time-ordered series of data to identify alterations in the underlying distributions, with the understanding that the sequence may contain multiple change points or none at all. Thus, the analysis serves a dual purpose: (a) determining the presence of any change points and (b) estimating their respective locations if they do exist.

This thesis focuses on conducting change-point analysis specifically for high-dimensional data. Our approach utilizes cluster analysis, wherein a suitable clustering algorithm is employed to partition the dataset into multiple clusters. Based on the outcomes of this clustering algorithm, we can identify the presence of change points. Recognizing that clustering based on Euclidean distance may not be optimal for non-convex clusters and can be challenging for high-dimensional data, especially when the underlying distributions exhibit varying dispersion structures beyond the first two moments, we will propose a novel change-point detection method. The effectiveness of this proposed method will be examined through theoretical and numerical investigations.

LIST OF SYMBOLS OR ABBREVIATIONS

∞	Infinity
$ \cdot $	Modulus
$\ \cdot\ $	Norm
WLLN	Weak Law of Large Numbers
MADD	Mean Absolute Difference of Distance
DI	DUNN's index
PDI	Penalized DUNN's index

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1. INTRODUCTION

Change point analysis plays a crucial role in various domains where understanding shifts or transitions in data patterns is essential. In high-dimensional datasets, where the number of variables or features is significantly large, detecting changes becomes even more challenging. This motivates the integration of clustering techniques into change point analysis to tackle the complexity of high-dimensional data.

One of the primary motivations for incorporating clustering in change point analysis of high-dimensional data is the ability to capture underlying structures and patterns. Clustering algorithms allow us to group similar data points together based on their feature values, thereby revealing hidden relationships and similarities. By applying clustering techniques to high-dimensional data, we can identify clusters that represent different data regimes or behaviours. These clusters can then be used as indicators of potential change points.

With the increasing prevalence of high-dimensional data sets in various fields, the analysis of complex and multivariate data has become a crucial area of research. High-dimensional data refers to data sets where the number of variables or features greatly exceeds the number of observations, presenting unique challenges for data analysis. Among the many tasks associated with high-dimensional data analysis, detecting changes or transitions in data patterns over time is of significant interest. This task, known as change point analysis, plays a vital role in numerous applications such as anomaly detection, quality control, and environmental monitoring.

Traditionally, change point analysis methods have been developed and widely used for low-dimensional data. However, applying these methods directly to high-dimensional data can be computationally demanding and may lead to inaccurate results due to the curse of dimensionality. To address these challenges, researchers have turned their attention to the integration of clustering techniques into change point analysis to leverage the inherent structure present in high-dimensional data.

Clustering, as a fundamental data mining technique, groups similar data points together based on their feature values. By applying clustering algorithms to high-dimensional data, we can identify clusters that represent distinct underlying patterns or behaviours. These clusters serve as a means to reduce the dimensionality of the data, effectively summarizing the behaviour of a group of variables and providing a more manageable representation for change point analysis.

Integrating clustering into change point analysis of high-dimensional data offers several advantages. First, it enables dimensionality reduction by grouping variables into clusters that exhibit similar patterns or behaviours. This reduction in dimensionality not only simplifies the analysis but also enhances interpretability by focusing on changes within clusters rather than individual variables.

Second, clustering-based change point analysis provides a robust approach to detecting significant transitions in high-dimensional data. By capturing the local structure within each cluster, clustering algorithms can effectively distinguish true changes from noise or fluctuations in the overall dataset. This ability to extract meaningful patterns within clusters improves the accuracy and reliability of change point detection in high-dimensional data.

Moreover, clustering-based change point analysis offers valuable insights into the dynamics and underlying structures of high-dimensional data. By identifying clusters that correspond to different data regimes or behaviours, we can gain a deeper understanding of how the data evolves over time. These clusters can help uncover hidden transitions, anomalies, or critical events that may have significant implications in various domains.

In this thesis, we aim to explore and develop novel approaches for change point analysis in high-dimensional data based on clustering techniques. We will investigate clustering algorithms with different dissimilarities and their integration with existing change point detection methods. Our objective is to propose efficient and effective methodologies that can handle the challenges posed by high-dimensional data, providing accurate detection of change points and valuable insights into the underlying dynamics of complex data sets.

Through extensive experiments and case studies using real-world data sets, we will demonstrate the applicability and performance of the proposed methods. The outcomes of this research will contribute to advancing change point analysis techniques tailored for high-dimensional data, enabling researchers and practitioners to uncover critical transi-

tions, detect anomalies, and make informed decisions based on the analysis of complex data sets.

Overall, by leveraging the power of clustering techniques in change point analysis, this research strives to address the unique challenges of high-dimensional data and contribute to the advancement of data analysis methodologies in diverse domains. In this thesis, we put forward an approach for change-point analysis that relies on clustering methods. What sets our proposed method apart is its suitability for High-Dimensional Low Sample Size (HDLSS) scenarios, where the number of variables is large while the available data points are limited.

The roots of change-point analysis can be traced back to statistical quality control, as highlighted by existing literature (Chen and Gupta, 2011[2]; Brodsky and Darkhovsky, 2013)[11]. These studies have made significant contributions to the field by presenting both parametric and nonparametric methods that excel in analyzing univariate data. In the realm of multivariate data, several noteworthy studies have contributed to change-point analysis within the parametric regime. Significant works in the area of change point analysis include [22, 26, 27, 23, 25]. These studies have proposed parametric methods that focus on change-point detection in multivariate datasets. Various nonparametric methods have been developed to detect changes in location, as evidenced by notable studies such as [3, 31, 24, 16]. These studies have contributed to the field by proposing effective techniques for identifying shifts in the central location of data without making specific distributional assumptions. The applicability of these methods extends to high-dimensional data, as evidenced by the works of [4, 30]. Various nonparametric methods have been developed to detect changes in distribution, offering viable options even in high-dimensional, low-sample size (HDLSS) scenarios. Notably, [23] devised graph-based two-sample tests, presenting innovative approaches within this domain. Various approaches have been explored for change-point analysis, encompassing tests based on minimum distance pairing [19], nearest neighbours ([13]), and minimum spanning tree [8]. [17] developed a method utilizing the energy statistic, as previously introduced in [28]. Furthermore, kernel-based methods have been discussed in [6, 12],

The organisation of the article is as follows. In chapter 2, we assume that the number of clusters is known. In subsection 2.1, we detect the change-point(s) using a traditional hierarchical clustering algorithm using Euclidean distance. However, this algorithm yields poor performance in some cases. Then, in subsection 2.3, we modified the traditional

hierarchical clustering algorithm. Still, it causes poor performance in some cases. So, in subsection 2.6, we propose a clustering approach that utilizes a dissimilarity measure derived from the data itself. In subsection 2.7, we investigate the high-dimensional behaviour of this dissimilarity. In the context where the dimensionality increases infinitely while the sample size remains fixed, the proposed methods demonstrate consistency under suitable regularity conditions. Specifically, under these conditions, the methods exhibit consistency when changes occur in terms of both location and scale.. We use a generalised dissimilarity function to detect changes outside the first two moments. Under appropriate regularity conditions, the high-dimensional consistency of the proposed methods is established in this thesis. In chapter 3, we estimate the optimal number of clusters. In subsection 3.2, we will show the performance of the algorithm and some simulated data sets. In subsection 3.3, we will analyse mathematically why our algorithm works properly in all kinds of data sets. In chapter 4, we will compare our algorithm with the state-of-the-art. In subsection 5, simulate our proposed method on the benchmark data set. This thesis comprehensively presents all the necessary proofs and mathematical intricacies associated with the research conducted.

2. DETECTION OF CHANGE-POINTS USING CLUSTERING

Change point analysis, also known as change detection or change point detection, refers to the process of identifying points or intervals in a sequence of data where the underlying statistical properties or behaviours significantly change. It is a fundamental task in various fields, including signal processing, time series analysis, quality control, finance, and environmental monitoring.

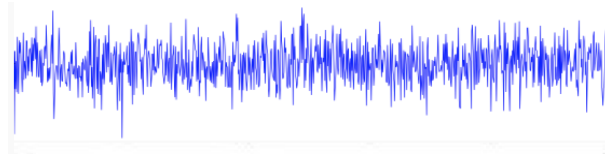


Fig. 2.1: No change point



Fig. 2.2: Single change point

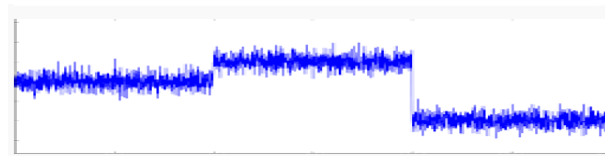


Fig. 2.3: Multiple change point

Example 1 : We consider an example with 60,2 observations, where first 20 observations follow $N(\mu_1, 0.5I)$, next 20 observations come from $N(\mu_2, 0.5I)$, and the remaining 20 come from $N(\mu_3, 0.5I)$, where $\mu_1 = [-2, 0]^T$, $\mu_2 = [2, 0]^T$, $\mu_3 = [0, 2\sqrt{3}]$, shown in the following figure [2.4](#)

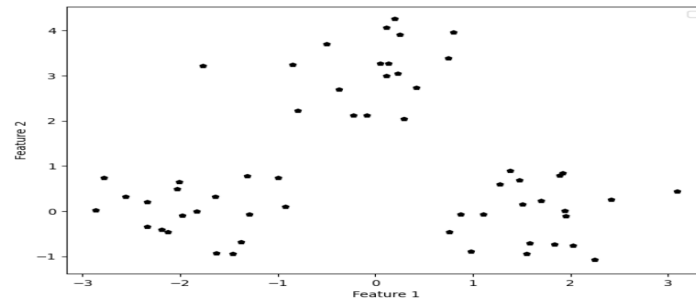


Fig. 2.4: Before clustering

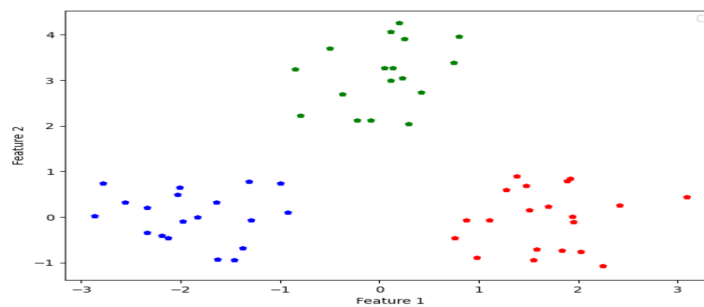


Fig. 2.5: After clustering:

2.1 What is Hierarchical clustering ?

Hierarchical clustering is a widely used technique for grouping objects based on their similarity. It organizes objects into clusters where the members within each cluster share similarities while being dissimilar to objects in other clusters. The resulting clusters are often depicted in a dendrogram, a hierarchical tree diagram that visually represents the relationships and structure of the clusters. This method allows for intuitive visualization and interpretation of the hierarchical relationships among the objects being clustered.

There are two main types of hierarchical clustering:

1. **Agglomerative:** The agglomerative hierarchical algorithm, a widely used clustering technique, follows a progressive merging process to create a hierarchical structure. Initially, each data point is treated as an individual cluster. The algorithm then iteratively combines the most similar clusters based on a chosen distance measure. This merging continues until all data points belong to a single cluster, resulting in a hierarchy of clusters with varying levels of similarity. The agglomerative hierarchical algorithm is known for its ability to capture nested relationships within the data and its flexibility in handling different distance metrics and linkage criteria.

2. **Divisive:** The Divisive method serves as the inverse of the Agglomerative method in hierarchical clustering. It begins with all objects grouped together in a single cluster and subsequently conducts a step-by-step splitting process until each object forms its own distinct cluster. The cluster division is guided by principles that emphasize maximizing the distance between neighboring objects within each cluster. This iterative splitting procedure allows for the creation of clusters that exhibit greater dissimilarity between their constituent objects

In this thesis, we prefer the Agglomerative method with a single linkage for detecting change points. This algorithm employs a measure of distance/similarity to create new clusters. Steps for Agglomerative clustering with single linkage can be summarized as follows:

Algorithm 1 Pseudo code for Hierarchical clustering:

Step 1: Compute the distance matrix using Euclidean distance

Step 2: Each data point is initially assigned to its own individual cluster.

Step 3: Merge the clusters based on the minimum distance between clusters

Step 4: Update the distance matrix based on single linkage.

Step 5: Repeat Step 3 and Step 4 until only a single cluster remains

Now applying *Hierarchical clustering* algorithm on **Example 1** in figure 2.5. Samples are in blue, red and green colours, representing they belong to cluster1, cluster2 and cluster3, respectively.

2.2 Does this method always work ?

Example 2 : Let us consider a problem, 60 observations $X_1, X_2, \dots, X_{60}$, where $X_1, X_2, \dots, X_{20} \sim N(\mu_1, 0.5I)$, $X_{21}, X_{22}, \dots, X_{40} \sim N(\mu_2, 0.5I)$, and $X_{41}, X_{42}, \dots, X_{60} \sim N(\mu_1, 0.5I)$, where $\mu_1 = [-2, 0]^T$, $\mu_2 = [2, 0]^T$

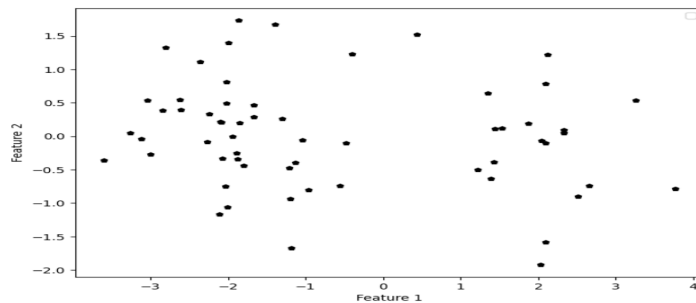


Fig. 2.6: Before clustering

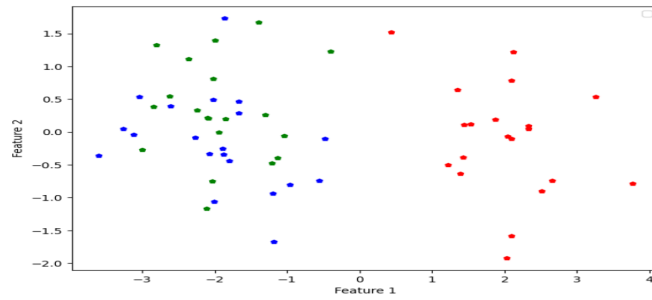


Fig. 2.7: After clustering

Performance : 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55
 56 57 58 59 60

So, we observe that the traditional *Hierarchical clustering* algorithm fails to detect the location of the change point properly.

2.3 How to take care of this?

We modify the traditional Hierarchical clustering algorithm to take care of the above problem. Now apply the modified Hierarchical clustering algorithm on **Example 2 , 3**

Algorithm 2 Pseudo code for modified Hierarchical clustering:

Step 1: Compute the distance matrix using Euclidean distance.

Step 2: Each data point is initially assigned to its own individual cluster..

Step 3: Merge the clusters based on the minimum distance between two consecutive clusters.

Step 4: Update the distance matrix based on single linkage.

Step 5: Repeat Step 3 and Step 4 until only a single cluster remains.

Performance on Example 2 :

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

Performance on Example 3 :

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

So, our modified algorithm works well in Example 2 and Example 3.

2.4 Does this method always works for high-dimensional data?

Let us consider the following three examples of 60 observations with dimension 100

Example 3 : $X_1, X_2, \dots, X_{60}$ observations of 100 dimension, where $X_1, X_2, \dots, X_{20} \sim N_{100}(\mu_1, 0.5I)$, $X_{21}, X_{22}, \dots, X_{40} \sim N(\mu_2, 0.5I)$, $X_{41}, X_{42}, \dots, X_{60} \sim N(\mu_3, 0.5I)$ and $\mu_1 = [0, 0, \dots, 0]^T$, $\mu_2 = [1, 1, \dots, 1]^T$, $\mu_3 = [-1, -1, \dots, -1]^T$

Performance:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

Example 4 : $X_1, X_2, \dots, X_{20} \sim N_{100}(\mu_1, 0.5I)$, $X_{21}, X_{22}, \dots, X_{40} \sim N(\mu_2, 0.5I)$, $X_{41}, X_{42}, \dots, X_{60} \sim N(\mu_1, 0.5I)$ and $\mu_1 = [-1, -1, \dots, -1]^T$ and $\mu_2 = [1, 1, \dots, 1]^T$

Performance:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

Example 5: $X_1, X_2, \dots, X_{30} \sim N_{100}(0, 2I)$ and $X_{31}, X_{32}, \dots, X_{60} \sim N_{100}(0, 4I)$

Performance:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

It worked well on **Example 4** and **Example 5** but didn't work on **Example 6**

2.5 Mathematical analysis of the reason for the failure

Let us consider $X_1, X_2, \dots, X_n$ n observations with d dimension where $X_1, X_2, \dots, X_\tau \sim N_d(0, \sigma_1^2 I)$, $X_{\tau+1}, X_{\tau+2}, \dots, X_n \sim N_d(\mu, \sigma_2^2 I)$ where, $\mu = (\mu, \mu, \dots, \mu)^T$.

For $1 \leq i, j \leq \tau$, by *Weak Law of Large number* (WLLN) we can show that as $d \rightarrow \infty$,

$$d^{-1} \|X_i - X_j\|^2 \xrightarrow{P} \begin{cases} 2\sigma_1^2 & \text{for } 1 \leq i, j \leq \tau \\ 2\sigma_2^2 & \text{for } \tau < i, j \leq n \\ \sigma_1^2 + \sigma_2^2 + \mu^2 & \text{for } i \leq \tau < j \text{ or } j \leq \tau < i \end{cases}$$

Proof. For $1 \leq i, j \leq \tau$, By WLLN as $d \rightarrow \infty$, $\frac{1}{d} \|X_i - X_j\|^2 \xrightarrow{P} E[\frac{1}{d} \|X_i - X_j\|^2]$. Now, $\frac{1}{d} \|X_i - X_j\|^2 = \frac{1}{d} \sum_{k=1}^d (x_{ik} - x_{jk})^2 = \frac{1}{d} \sum_{k=1}^d w_k^2$, where $w_k = x_{ik} - x_{jk} \sim N(0, 2\sigma_1^2)$, $\text{var}(w_k) = \text{var}(x_{ik}) + \text{var}(x_{jk}) - 2\text{cov}(x_{ik}, x_{jk}) = \sigma_1^2 + \sigma_1^2 - 0 = 2\sigma_1^2$, $E(w_k^2) = \text{var}(w_k) + \{E(w_k)\}^2$. So, $E[\frac{1}{d} \|X_i - X_j\|^2] =$

$2\sigma_1^2$. Therefore, $\frac{1}{d}\|X_i - X_j\|^2 \rightarrow 2\sigma_1^2$. Similarly, for $(\tau + 1) \leq i, j \leq n$, $\frac{1}{d}\|X_i - X_j\|^2 \rightarrow 2\sigma_2^2$ and for $1 \leq i \leq \tau, \tau + 1 \leq j \leq n$ or $1 \leq j \leq \tau, \tau + 1 \leq i \leq n$, $\frac{1}{d}\|X_i - X_j\|^2 = \frac{1}{d} \sum_{k=1}^d (x_{ik} - x_{jk})^2 = \frac{1}{d} \sum_{k=1}^d b_k^2$, where $b_k = x_{ik} - x_{jk} \sim N(-\mu, \sigma_1^2 + \sigma^2)$, $\text{var}(w_k) = \text{var}(x_{ik}) + \text{var}(x_{jk}) - 2\text{cov}(x_{ik}, x_{jk}) = \sigma_1^2 + \sigma_2^2 - 0 = \sigma_1^2 + \sigma_2^2$, $E(b_k^2) = \text{var}(b_k) + \{E(b_k)\}^2 = \sigma_1^2 + \sigma_2 + \mu^2$. Let us define $d_{11} = 2\sigma_1^2$, $d_{22} = 2\sigma_2^2$ and $d_{12} = \sigma_1^2 + \sigma_2^2 + \mu^2$.

Location Problem : In examples 4 and 5, $d_{11} = d_{22} < d_{12}$, i.e. the distance between two observations from the same distribution is smaller than the distance between two observations from the different distributions.

Scale Problem: In example 5, $d_{11} < d_{12} < d_{22}$, i.e. the distance between two observations from the same distribution is larger than the distance between two observations from the different distributions.

2.6 What is the remedy?

Euclidean distance does not work properly in the data set with different dispersion structures. So, we use a new dissimilarity measure called Mean Absolute Difference of Distance (MADD).

MADD:[21] For a set consisting of n observations $X_1, X_2, \dots, X_n$, MADD between two observations X_i and X_j as defined as

$$\delta_0(X_i, X_j) = \frac{1}{n-2} \sum_{k=1, k \neq i, j}^n \left| \|x_i - x_k\| - \|x_j - x_k\| \right|$$

2.7 High-dimensional behaviour of MADD

If $X_1, X_2, \dots, X_\tau \sim N_d(0, \sigma_1^2 I)$ and $X_{\tau+1}, X_{\tau+2}, \dots, X_n \sim N_d(\mu, \sigma_2^2 I)$, where $\mu = (\mu, \mu, \dots, \mu)^T$

If $X_1, X_2, \dots, X_\tau \sim N_d(0, \sigma_1^2 I)$ and $X_{\tau+1}, X_{\tau+2}, \dots, X_n \sim N_d(\mu, \sigma_2^2 I)$, where $\mu = (\mu, \mu, \dots, \mu)^T$. By Weak Law of Large number (WLLN), we can show that as $d \rightarrow \infty$,

$$d^{-1/2}(n-2)\delta_1(X_i, X_j) \xrightarrow{P} \begin{cases} 0 & \text{for } 1 \leq i, j \leq \tau \\ 0 & \text{for } \tau < i, j \leq n \\ d_{12}^* & \text{for } i \leq \tau < j \text{ or } j \leq \tau < i, \end{cases}$$

$$\text{where } d_{12}^* = (\tau - 1) \left| \sqrt{\mu^2 + \sigma_1^2 + \sigma_2^2} - \sigma_1 \sqrt{2} \right| + (n - \tau - 1) \left| \sqrt{\mu^2 + \sigma_1^2 + \sigma_2^2} - \sigma_2 \sqrt{2} \right|$$

Proof. For $1 \leq i, j \leq \tau$, $d^{-1/2}(n-2)\delta(X_i, X_j) = d^{-1/2} \left[\sum_{1 \leq k \leq \tau, k \neq i, j} \left| \|X_i - X_k\| - \|X_j - X_k\| \right| + \sum_{\tau+1 \leq k \leq n, k \neq i, j} \left| \|X_i - X_k\| - \|X_j - X_k\| \right| \right]$. Now, for $1 \leq k \leq \tau$, by WLLN, $d^{-1/2} \|X_i - X_k\| \xrightarrow{P}$

$\sqrt{2}\sigma_1$. Similarly, $d^{-1/2} \|X_i - X_k\| \xrightarrow{p} \sqrt{2}\sigma_1$. So, $d^{-1/2} \sum_{1 \leq k \leq \tau, k \neq i, j} |\|X_i - X_k\| - \|X_k - X_j\|| \xrightarrow{p} 0$. Also, for $\tau + 1 \leq k \leq n$, $d^{-1/2} \|X_i - X_k\| \xrightarrow{p} \sqrt{\sigma_1^2 + \sigma_2^2}$ and $d^{-1/2} \|X_k - X_j\| \xrightarrow{p} \sqrt{\sigma_1^2 + \sigma_2^2}$. So, $d^{-1/2} \sum_{1 \leq k \leq \tau, k \neq i, j} |\|X_i - X_k\| - \|X_k - X_j\|| \xrightarrow{p} 0$. Therefore, for $1 \leq i, j \leq \tau$, $d^{-1/2}(n-2)\delta(X_i, X_j) \xrightarrow{p} 0$. Similarly, $\tau + 1 \leq i, j \leq n$, $d^{-1/2}(n-2)\delta(X_i, X_j) \xrightarrow{p} 0$. Now, in a similar way for $i \leq \tau \leq j$ or $j \leq \tau \leq i$, $d^{-1/2}(n-2)\delta(X_i, X_j) \xrightarrow{p} (\tau - 1)|\sqrt{\mu^2 + \sigma_1^2 + \sigma_2^2} - \sigma_1\sqrt{2}| + (n - \tau - 1)|\sqrt{\mu^2 + \sigma_1^2 + \sigma_2^2} - \sigma_2\sqrt{2}|$. The expression $d_{12}^* = 0$ iff $\sigma_1^2 = \sigma_2^2$. If $\sigma_1^2 \neq \sigma_2^2$, we have $d_{12}^* > 0$. Therefore, we conclude that in both location and scale problems, the MADD observations from the same distribution are similar to the MADD between two observations from different distributions.

2.8 What if $\mu^2 = 0$ and $\sigma_1^2 = \sigma_2^2$?

Example 6: Let, $X_1, X_2, \dots, X_{30} \sim t_4(0, I)$ and $X_{31}, X_{32}, \dots, X_{60} \sim N_{100}(0, 2I)$.

Performance:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

Our proposed algorithm failed to detect the proper location of the change point in the above example.

2.9 How to overcome this limitation?

Now we introduce the MADD-based l_1 [10] norm dissimilarity. For a set consisting of n observations $X_1, X_2, \dots, X_n$, the l_1 distance based MADD between X_i and X_j is defined as,

$$\delta_1(X_i, X_j) = \frac{1}{n-2} \sum_{k=1, k \neq i, j}^n \left| \sum_{l=1}^d |X_i^{(l)} - X_k^{(l)}| - \sum_{l=1}^d |X_j^{(l)} - X_k^{(l)}| \right|$$

Now apply the algorithm based on the above distance in **Example 7**

Performance:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

So, it worked well in the **Example 6**

2.10 High dimensional behaviour of MADD based on l_1 dissimilarity

Let $X_1, X_2, \dots, X_\tau \stackrel{\text{iid}}{\sim} F$ and $X_{\tau+1}, X_{\tau+2}, \dots, X_n \stackrel{\text{iid}}{\sim} G$

Under some appropriate assumptions $\star\star$,

$$d^{-1}(n-2)\delta_1(X_i, X_j) \xrightarrow{P} \begin{cases} 0 & \text{for } 1 \leq i, j \leq \tau \\ 0 & \text{for } \tau < i, j \leq n \\ d_{12}^{\star\star} & \text{for } i \leq \tau < j \text{ or } j \leq \tau < i, \end{cases}$$

where $d_{12}^{\star\star} \geq 0$ and the equality holds if and only if F and G have identical marginal distributions.

Proof. To prove this, we use the following lemma

Let X_1, X_2, Y_1, Y_2 be independent real random variables with finite expectations. Let X_1, X_2 be identically distributed with distribution function F and let Y_1, Y_2 be identically distributed with distribution function G [5]. Then

$$2E \|X_1 - Y_1\| - E \|X_1 - X_2\| + E \|Y_1 - Y_2\| \geq 0$$

where the equality holds if and only if $F = G$

Here, we use a one-dimensional version of the above expression

Let $X_1^{(i)}, X_2^{(i)} \sim F^{(i)}$ and $Y_1^{(i)}, Y_2^{(i)} \sim G^{(i)}$

$$2E|X_1^{(i)} - Y_1^{(i)}| - E|X_1^{(i)} - X_1^{(i)}| - E|Y_1^{(i)} - Y_2^{(i)}| \geq 0$$

the equality holds if and only if $F^{(i)} = G^{(i)}$

So, $\delta_1(X_l, X_k) = \frac{1}{n-2} \sum_{j=l, k \neq i, j}^n \left| \sum_{i=1}^d |X_l^{(i)} - X_j^{(i)}| - \sum_{i=1}^d |X_k^{(i)} - X_j^{(i)}| \right|$, where $1 \leq l, k \leq n$

$$\text{or, } \frac{\delta_1(X_l, X_k)}{d} = \frac{1}{d(n-2)} \sum_{j=l, k \neq i, j}^n \left| \sum_{i=1}^d |X_l^{(i)} - X_j^{(i)}| - \sum_{i=1}^d |X_k^{(i)} - X_j^{(i)}| \right|$$

Now, by WLLN $\frac{\delta_1(X_l, X_k)}{d} \xrightarrow{P} \frac{1}{(n-2)} \left[\sum_{j=l, k \neq i, j}^n \left| \frac{1}{d} \sum_{i=1}^d E|X_l^{(i)} - X_j^{(i)}| - \frac{1}{d} \sum_{i=1}^d E|X_k^{(i)} - X_j^{(i)}| \right| \right]$

If X_l and X_k come from the same population, $E|X_l^{(i)} - X_j^{(i)}| = E|X_k^{(i)} - X_j^{(i)}| = 0$, for $1 \leq l \leq \tau$ and $\tau + 1 \leq k \leq n$, $\frac{1}{(n-2)} \left[\sum_{j=l, k \neq i, j}^n \left| \frac{1}{d} \sum_{i=1}^d E|X_l^{(i)} - X_j^{(i)}| - \frac{1}{d} \sum_{i=1}^d E|X_k^{(i)} - X_j^{(i)}| \right| \right] \geq \frac{1}{(n-2)} \left[\left| \frac{1}{d} \sum_{i=1}^d E|X_l^{(i)} - X_2^{(i)}| - \frac{1}{d} \sum_{i=1}^d E|X_k^{(i)} - X_2^{(i)}| \right| + \left| \frac{1}{d} \sum_{i=1}^d E|X_l^{(i)} - X_n^{(i)}| - \frac{1}{d} \sum_{i=1}^d E|X_k^{(i)} - X_n^{(i)}| \right| \right] \geq \frac{1}{n-2} \left[\frac{1}{d} \sum_{i=1}^d E|X_l^{(i)} - X_2^{(i)}| - \frac{1}{d} \sum_{i=1}^d E|X_k^{(i)} - X_2^{(i)}| + \frac{1}{d} \sum_{i=1}^d E|X_l^{(i)} - X_n^{(i)}| - \frac{1}{d} \sum_{i=1}^d E|X_k^{(i)} - X_n^{(i)}| \right] \geq \frac{1}{d(n-2)} \sum_{i=1}^d [2E|X_l^{(i)} - X_k^{(i)}| - E|X_l^{(i)} - X_2^{(i)}| + E|X_k^{(i)} - X_2^{(i)}|]$

$|X_n^{(i)}| \geq 0$ (by the above lemma). Equality holds when the marginal distribution is the same for both populations.

Assumptions★(Verbose from[21]): We assume that, $X_1, X_2, \dots, X_n$ are n independent observations d dimensional random variables coming from k_0 populations $F_1, F_2, \dots, F_{k_0}$. Let μ_i and $\sum_i$ denote the mean vector and the dispersion matrix of F_i , respectively. We also make the following assumptions.

(A1) The fourth moments of the measurement variables are uniformly bounded.

(A2) For two independent observation $U \sim F_i$ and $V \sim F_j, \sum_{q \neq q'} \text{cov}\{(U^{(q)} - V^{(q)})^2, (U^{(q')} - V^{(q')})^2\} = \mathbf{o}(d^2)$

(A3) There exist finite quantities ν_{ij} and σ_i^2 such that $d^{-1}\|\mu_i - \mu_j\|^2 \rightarrow \nu_{ij}$ and $d^{-1}\text{trace}(\sum_i) \rightarrow \sigma_i^2$ as $d \rightarrow \infty$

3. DETERMINING THE COUNT OF CHANGE POINTS: ESTIMATION AND ANALYSIS

Previously, we have been presuming that the number of change points is a known quantity. However, in practical applications, it becomes necessary to estimate this number from the given data set. This particular challenge is more complex than simply identifying the locations of the change points. In the context of this thesis, our approach involves estimating the number of change points by determining the number of clusters. Assuming we have k clusters, it follows that there would be $k-1$ change points under the assumption that at least one change point exists within the data set. Various estimation methods exist to determine the number of clusters., such as *KL statistic*[15], *Gap statistic*[29], *JUMP statistics*[14], *Cross-validation Rand index*, *Silhouettes coefficient*[20], *DUNN index*[9] . All the existing methods follow a general strategy. In this thesis, we use a modified DUNN index to estimate the number of change points.

3.1 DUNN index

For a fixed k , let $C_1, C_2, \dots, C_K$ be the clusters estimated by our proposed algorithm. Define *within cluster distance* ($MADD l_1$) for i th cluster C_i ,

$$W_K(i) = \frac{1}{|C_i|(|C_i| - 1)} \sum_{X_r \in C_i} \sum_{X_s \in C_i, r \neq s} \delta_1(X_r, X_s)$$

and *between cluster distance* between C_i and C_j is defined as,

$$B_K(i, j) = \frac{1}{|C_i||C_j|} \sum_{X_r \in C_i} \sum_{X_s \in C_j} \delta_1(X_r, X_s)$$

where $|C|$ denotes the number of samples in cluster C

Here, $\binom{k}{2}$ many $B_K(i, j)$ ($1 \leq i < j \leq K$) values and k many $W_K(i)$ ($1 \leq i \leq K$) values.

Now define $W(K) = \max_{1 \leq i \leq K} W_K(i)$ and $B(K) = \min_{1 \leq i < j \leq K, i-j=1} B_K(i, j)$. Now the *DUNN index* is defined by,

$$D(K) = \frac{B(K)}{W(K)}$$

Here, we use it to estimate the number of clusters by $\hat{K}_D = \operatorname{argmax}_{2 \leq k \leq K} D(k)$. We apply our proposed algorithm with MADD l_1 dissimilarity on all the previous examples 100 times to detect the number of change points. The following table shows bar plots for the number of clusters estimated by our proposed method in **Examples 1-6**.

3.2 Performance of our proposed method based on DUNN index

	k = 2	k = 3	k = 4	k = 5	k ≥ 6
Example1	0	100	0	0	0
Example2	5	85	3	4	3
Example3	0	100	0	0	0
Example4	0	100	0	0	0
Example5	0	100	0	0	0
Example6	40	50	0	0	0

Tab. 3.1: Number of clusters estimated by DUNN index in **Example 1-6**

So, we observe that our modified DUNN index performs outstandingly in all the **Example 1-5** except **Example 6**.

3.3 Theoretical investigation of modified Dunn index on high-dimensional data

The success of modifying Dunn's index in all the examples inspired us to investigate its theoretical aspect. We aim to examine and analyze the behaviour of the Dunn index by leveraging insights and methodologies from existing literature and research papers with modification. For this investigation, we make some assumptions on the asymptotic orders of MADD l_1 dissimilarity (Verbose from [21]). For the observations $X \stackrel{\text{iid}}{\sim} F_i$ and $Y \stackrel{\text{iid}}{\sim} F_j$. Let as $d \rightarrow \infty$, $\delta_1(X, Y) \xrightarrow{p} \psi_{ij}(d)$, i.e. $\Pr\{\delta_1(X, Y)/\psi_{ij}(d) \text{ bounded away from 0 and } \infty\} \rightarrow 1$. Assume that, within cluster separations are of the same asymptotic order, and so are all between cluster separations, i.e.

$\psi_{ii}(d) \rightarrow \psi_-(d)$ for all $i = 1, 2, \dots, K$ and $\psi_{ij}(d) \rightarrow \psi_+(d)$ for $1 \leq i \neq j \leq K$, where $\psi_-(d) = o(\psi_+(d))$ and $\psi_-(d) = o(d^{-\alpha})$, where some $\alpha > 0 \dots \dots \dots (A1)$

We will elegantly demonstrate that the given theorem holds true even when applied to change point data.

Theorem 3.1. *Suppose that we have observations from $K \geq 2$ populations which satisfy (A1), and also assume that K_{opt} is the optimal number of clusters of the base clustering algorithm. Then as $d \rightarrow \infty$, $\hat{K}_D \xrightarrow{p} K$ ($\hat{K}_D$ is the number of clusters estimated by MADD l_1 version) [21]*

Proof. *Case(1): For any $k < K_{opt}$,*

Claim(1): *There is at least one cluster that includes observations from two distinct populations*

Claim(2): *No two clusters contain observations from the same population.*

From claim(1) and under the assumptions A1, we get, $W_k(i) \xrightarrow{p} \psi_+(d)$, for some i . Hence $W(k) = \max_{1 \leq i \leq k} W_i(k) \xrightarrow{p} \psi_+(d)$. Also, by claim(2) and under A1, $B_k(i, j) \xrightarrow{p} \psi_+(d)$ for every $i \neq j$. Hence, $B(k) = \min_{1 \leq i < j \leq k} B_k(i, j) \xrightarrow{p} \psi_+(d)$. So, $D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} C_1$ (some constant).

Case(2): For any $k > K_{opt}$,

Claim(1): *No cluster contains observations from different populations.*

Claim(2): *At least two clusters contain observations from the same populations.*

From claim(1) and under the assumptions A1, we get, $B_k(i, j) \xrightarrow{p} \psi_-(d)$, for some $i \neq j$. Hence $B(k) = \min_{1 \leq i < j \leq k} B_k(i, j) \xrightarrow{p} \psi_-(d)$. Also, by claim(2) and under A1, $W_k(i) \xrightarrow{p} \psi_-(d)$ for every i . Hence, $W(k) = \max_{1 \leq i \leq k} W_k(i) \xrightarrow{p} \psi_-(d)$. So, $D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} C_2$ (some constant).

Case(3): For any $k = K_{opt}$,

Claim(1): Each cluster contains observations from same population.

Claim(2): Two different clusters contain observations from two different populations.

From claim(1) and under A1, we get, $W_k(i) \xrightarrow{p} \psi_-(d)$, for every i . Hence $W(k) = \max_{1 \leq i \leq k} W_i(k) \xrightarrow{p} \psi_-(d)$. Also, by claim(2) and under A1, $B_k(i, j) \xrightarrow{p} \psi_+(d)$ for every $i \neq j$. Hence, $B(k) = \min_{1 \leq i < j \leq k} B_k(i, j) \xrightarrow{p} \psi_+(d)$. So, $D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} \infty$.

Now, $\Pr(D(k_{opt}) > D(k) \forall k \neq k_{opt}) \xrightarrow{p} 1 \implies \hat{K}_D \xrightarrow{p} k_{opt}$ as $d \rightarrow \infty$

3.4 Limitation of modified *DUNN*'s index

Example 7 : Let $X_1, X_2, \dots, X_{60} \sim N_{100}(0, 0.5I)$. Perform modified *DUNN*'s index on this example.

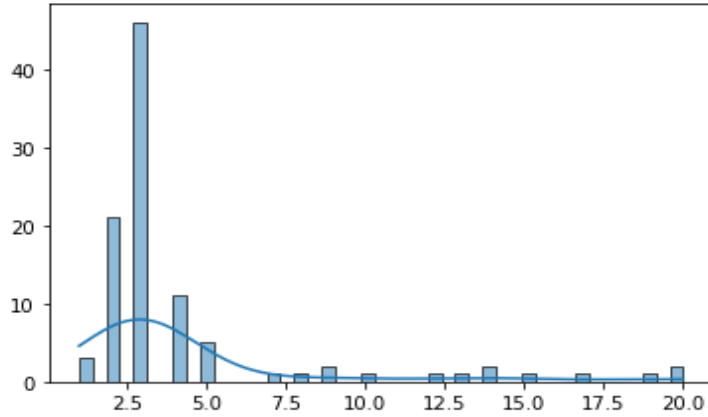


Fig. 3.1: Example 7

So, it demonstrates poor performance in **Example 6**. Therefore, the limitation of the Dunn index is its inability to accurately detect the number of clusters when only a single cluster exists in the data set. Since the Dunn index evaluates the compactness and separation of clusters, it heavily relies on the presence of multiple distinct clusters for meaningful analysis. In the case of a single cluster, the index fails to provide informative insights regarding the optimal number of clusters. Consequently, researchers and practitioners should exercise caution when relying solely on the Dunn index in scenarios where the data set consists of a single dominant cluster, as it may lead to misleading interpretations and inappropriate cluster selection.

Mathematical reason behind this : For any $k = K_{opt}$, under A1, we get, $W_k(i) \xrightarrow{p} \psi_-(d)$, for every i . Hence $W(k) = \max_{1 \leq i \leq k} W_i(k) \xrightarrow{p} \psi_-(d)$. Also, $B_k(i, j) \xrightarrow{p} \psi_-(d)$ for every $i \neq j$. Hence, $B(k) = \min_{1 \leq i < j \leq k} B_k(i, j) \xrightarrow{p} \psi_-(d)$. So, $D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} C$. (some constant) $\not\Rightarrow Pr(D(k_{opt}) > D(k) \forall k \neq k_{opt}) \xrightarrow{p} 1$, i.e. $\hat{K}_D \xrightarrow{p} k_{opt}$ as $d \rightarrow \infty$

3.5 How to overcome this limitation ?

Complementary techniques or alternative clustering metrics should be considered to overcome this particular limitation and ensure the accurate characterization of data sets with single-cluster structures. To mitigate this limitation, we define a modified **penalized version of DUNN index** (PD)[21]. For any fixed k , it is defined as $PD(k) = \frac{B(k)}{W(k)} - k\zeta(d)$ and $W(k)$ has the same sense as in the DUNN's index and $B(1) = B(2)$, $\zeta(d)$ is the penalty function.

Theorem 3.2. Suppose we have observations from $k_{opt} \geq 1$ population(s) which satisfies the assumptions A1 and the penalized function $\zeta(d)$ is such that $\zeta(d) \rightarrow \infty$ and $\frac{\psi_-(d)\zeta(d)}{\psi_+(d)} \rightarrow 0$ as $d \rightarrow \infty$. If k_{opt} is the optimal number of clusters, then $\hat{K}_{PD} \xrightarrow{p} k_{opt}$ [21].

Proof. Case(1): For any $k \geq 2$,

$$D(k) \xrightarrow{p} \begin{cases} \frac{\psi_+(d)}{\psi_-(d)} & \text{for } k = k_{opt} \\ C & \text{for } k \neq k_{opt} \end{cases}$$

where C is constant

For $k \neq k_{opt}$, $PD(k_{opt}) - PD(k) = D(k_{opt}) - k_{opt}\zeta(d) - D(k) + k\zeta(d) = D(k_{opt}) - D(k) - (k_{opt} - k)\zeta(d) \xrightarrow{p} (\frac{\psi_+(d)}{\psi_-(d)} - C) - (k_{opt} - k)\zeta(d) \xrightarrow{p} \infty$ (Since, $\zeta(d) = o(\frac{\psi_+(d)}{\psi_-(d)})$ as $d \rightarrow \infty$). Thus, $\hat{K}_{PD} \xrightarrow{p} k_{opt}$

Case(2): When $k_{opt} = 1$, we have $W(k) \xrightarrow{p} \psi_-(d)$, $B(k) \xrightarrow{p} \psi_-(d)$ for all $k \geq 1$. Therefore, $D(k) \xrightarrow{p} C$. For $k \neq 1$, $PD(1) - PD(k) = D(1) - \zeta(d) - D(k) + k\zeta(d) = D(1) - D(k) + (k - 1)\zeta(d) \xrightarrow{p} \infty$, as $d \rightarrow \infty$. Hence $\hat{K}_{PD} \xrightarrow{p} \infty$ as $d \rightarrow \infty$

Therefore, the penalized Dunn index is capable of detecting scenarios where there is only one cluster present. As a measure of cluster compactness and inter-cluster dissimilarity, the penalized Dunn index quantifies the quality of clustering. In the case of a single cluster, the data points are expected to be highly compact and exhibit minimal dissimilarity, resulting in a higher penalized Dunn index value. Thus, when there is only one cluster present, the penalized Dunn index can effectively capture this homogeneous structure and signal the absence of distinct clusters or change points in the data. Now, perform modified penalized DUNN's index in **Example 1-6**

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k \geq 6$	Total
Example1	0	0	100	0	0	0	100
Example2	0	0	100	0	0	0	100
Example3	0	0	100	0	0	0	100
Example4	0	0	100	0	0	0	100
Example5	0	0	100	0	0	0	100
Example6	10	50	40	0	0	0	90
Example7	100	0	0	0	0	0	100

Tab. 3.2: Number of clusters estimated by Penalized DUNN index in **Example 1-7**

Note: Total represents the number of times the change point is found. Bold figures indicate the best result in each example.

So, it performs well in all examples except **Example 6**

4. EXPLORING SIMULATION AND BENCHMARK DATASETS: AN ANALYTICAL INVESTIGATION

In this section, we conduct a comparative analysis of our proposed methods with several state-of-the-art techniques, utilizing additional data sets. Specifically, we evaluate the performance against the E-divisive method utilizing energy distances[18, 12, 1] and the graph-based methods introduced in [7]. Within the realm of graph-based methods, our focus lies on exploring approaches based on the minimum spanning tree (MST), nearest neighbour graph (NNG), and minimum distance pairing (MDP). It is important to highlight that the efficacy of the kernel method is contingent upon several factors.

We have implemented our proposed methods python3 codes, and they are accessible in this link.

4.1 Result of the simulated data set

We conduct experiments with a total of six examples, each consisting of 40 observations obtained from 250-dimensional distributions. In each example, the initial τ samples are coming from one distribution F_1 , while the remaining observations are generated from another distribution G_2 . We explore three different values for τ (10, 20, and 30), and for each value, we repeat the experiment 100 times. The outcomes and effectiveness of various methods are consolidated and presented in the summarized Table.

In the initial two examples, our focus is on two multivariate normal distributions that exhibit variations in terms of location and scales

. **Example A**, The distributions $F_1 = N_{250}(0_{250}, \Sigma_0)$ and $F_2 = N_{250}(1_{250}, \Sigma_0)$ have identical scatter matrices $\Sigma_0 = ((0.9^{|i-j|}))_{250 \times 250}$, but distinct mean vectors. For this location-based problem, all methods, except MDP, performed admirably. DI, PDI and ker-

nel methods exhibited the highest performance, closely followed by the E-divisive method. The MST method, however, demonstrated relatively lower detection rates.

Example B, The distributions $F_1 = N_{250}(0_{250}, \sum_0)$ and $F_2 = N_{250}(0_{250}, 3\sum_0)$ solely differ in their scatter matrices, where $\sum_0$ is same as **Example A**. Within the context of this scaling problem, our suggested approaches, specifically DI and PDI, showcased superior performance compared to their counterparts. Out of the alternative methods employed, only the kernel method exhibited a moderately competitive performance. However, it is worth noting that the techniques relying on MST and NNG consistently failed to identify the accurate change point.

Example C : The distributions $F_1 = N_{200}(0_{200}, \sum_1)$ and $F_2 = N_{200}(0_{200}, \sum_2)$. Here, $\sum_1$, the first 100 diagonal elements are 1, and the rest are 3 and $\sum_2$, the first 100 diagonal elements are 3, and the rest are 1

Example D : $F_1 = N_{200}(0_{200}, 2I)$ and $F_2 = t_4(0, I)$, the mean vector and the dispersion matrix are the same for both the distributions.

		$\tau = 10$							$\tau = 20$							$\tau = 30$										
$ \hat{\tau} - \tau $		0	1	2	3	4	5	≥ 6	Total	0	1	2	3	4	5	≥ 6	Total	0	1	2	3	4	5	≥ 6	Total	
Example A	DI	92	3	0	0	0	0	0	95	100	0	0	0	0	0	0	100	91	3	0	0	0	0	0	0	94
	PDI	88	0	3	0	0	0	0	91	98	2	0	0	0	0	0	100	83	0	0	0	0	0	0	7	90
	E-div	90	9	0	0	0	0	1	100	86	8	3	0	0	0	3	100	84	6	1	1	0	0	0	8	100
	Kernel	94	6	0	0	0	0	0	100	88	9	3	0	0	0	0	100	91	8	1	0	0	0	0	0	100
	MST	68	10	8	1	4	1	7	99	66	18	6	2	0	1	7	100	64	11	2	6	1	1	11	96	
	NNG	84	11	2	1	1	0	0	100	83	12	5	0	0	0	0	100	81	12	4	2	1	0	0	0	100
	MDP	44	6	6	6	0	5	30	97	51	13	5	1	2	0	22	94	37	15	5	3	3	1	26	90	
Example B	DI	90	0	0	0	0	5	0	95	100	0	0	0	0	0	0	100	92	0	0	0	0	2	5	99	
	PDI	90	0	0	0	0	5	0	95	100	0	0	0	0	0	0	100	92	0	0	0	0	2	5	99	
	E-div	10	4	2	0	1	1	7	25	41	16	7	5	2	0	7	78	27	16	6	3	1	1	3	57	
	Kernel	54	14	7	3	3	3	16	100	82	10	5	3	0	0	0	100	89	10	1	0	0	0	0	100	
	MST	0	0	0	0	1	3	28	32	0	0	0	0	0	0	17	17	0	0	0	0	0	0	14	14	
	NNG	0	0	0	0	0	0	3	3	0	0	0	0	0	0	4	4	0	0	0	0	0	0	3	3	
	MDP	30	10	7	0	4	10	25	86	36	6	2	1	0	2	21	68	16	7	4	3	2	1	30	63	
Example C	DI	100	0	0	0	0	0	0	100	100	0	0	0	0	0	0	100	100	0	0	0	0	0	0	0	100
	PDI	100	0	0	0	0	0	0	100	100	0	0	0	0	0	0	100	100	0	0	0	0	0	0	0	100
	E-div	0	0	0	1	0	0	8	9	1	1	0	0	0	1	9	12	0	0	0	1	0	0	2	3	
	Kernel	2	7	7	6	4	3	71	100	1	5	2	6	1	3	82	100	3	6	3	2	2	8	75	99	
	MST	1	2	2	3	3	0	12	23	2	2	2	1	1	0	22	30	2	4	1	3	0	1	21	32	
	NNG	2	1	4	4	2	0	8	21	5	5	2	0	2	1	19	34	2	3	2	2	0	3	8	20	
	MDP	4	3	0	2	1	3	26	39	3	2	1	0	1	0	25	32	5	1	1	1	2	4	21	35	
Example D	DI	35	0	0	0	0	0	45	80	48	0	0	0	0	4	37	89	34	0	0	0	0	0	48	82	
	PDI	30	0	0	0	0	12	40	82	43	0	5	6	5	0	27	87	30	0	3	4	2	5	46	90	
	E-div	0	0	0	0	0	0	6	6	2	0	0	1	0	0	2	5	0	1	0	0	0	0	3	4	
	Kernel	0	5	8	4	2	4	76	99	2	3	3	3	2	5	82	100	3	3	5	6	7	2	71	97	
	MST	0	0	0	0	0	0	9	9	0	0	0	0	0	0	20	20	0	0	1	0	1	1	17	20	
	NNG	0	0	0	0	0	0	12	12	0	0	0	0	0	0	9	9	1	1	1	0	1	0	11	15	
	MDP	0	0	0	0	0	0	12	12	0	0	0	0	0	0	18	18	0	0	0	0	0	0	13	13	

Tab. 4.1: Frequency distribution of $|\hat{\tau} - \tau|$, where $\hat{\tau}$ is the predicted location of change point

In Example C and D The distributions under consideration share identical location and dispersion matrix trace, yet they exhibit variations in their marginal distributions. In such instances, DI and PDI methods demonstrate excellent performance in accurately discriminating them in 100 iterations. Conversely, the remaining competing techniques display subpar performance. This outcome was anticipated, given that these competing methods rely on pairwise Euclidean distances, which are unsuitable for capturing the dis-

tinguishing characteristics of the distributions in question.

4.2 Multiple change point detection

Let us consider the following two examples :

Example 12: $X_1, X_2, \dots, X_{15} \sim N_{250}(0.5\mathbf{1}_{250}, \mathbf{I}_{250})$,

$X_{16}, X_{17}, \dots, X_{30} \sim N_{250}(0_{250}, \mathbf{I}_{250})$,

$X_{31}, X_{32}, \dots, X_{45} \sim N_{250}(0.5\mathbf{1}_{250}, \mathbf{I}_{250})$,

Example 13: $X_1, X_2, \dots, X_{60}$ observations from four Gaussian distributions $N_{250}(0_{250}, (\frac{1}{20})^{i-1}\mathbf{I}_{250})$, $i = 1, 2, 3, 4$, differing only their scales. The first 15 observations are generated from first distribution $N_{250}(0_{250}, \mathbf{I}_{250})$, next 15 from the second and so on .

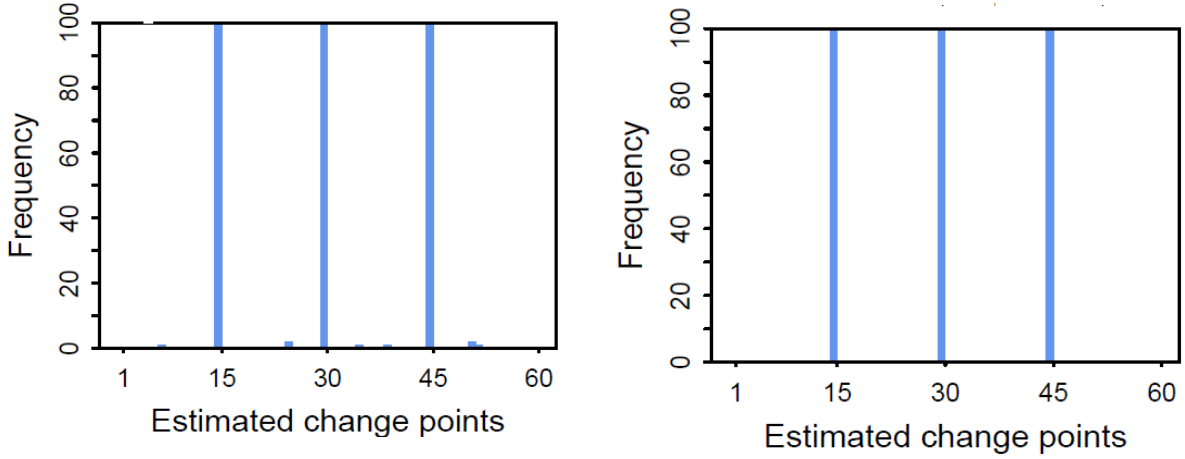


Fig. 4.1: E-div and Kernel method in example 12

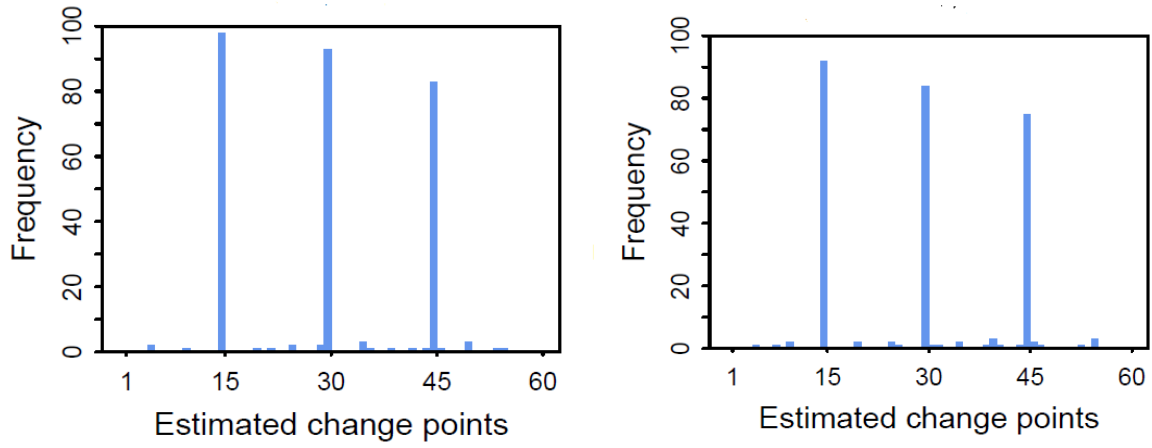


Fig. 4.2: DUNN index and Penalized index

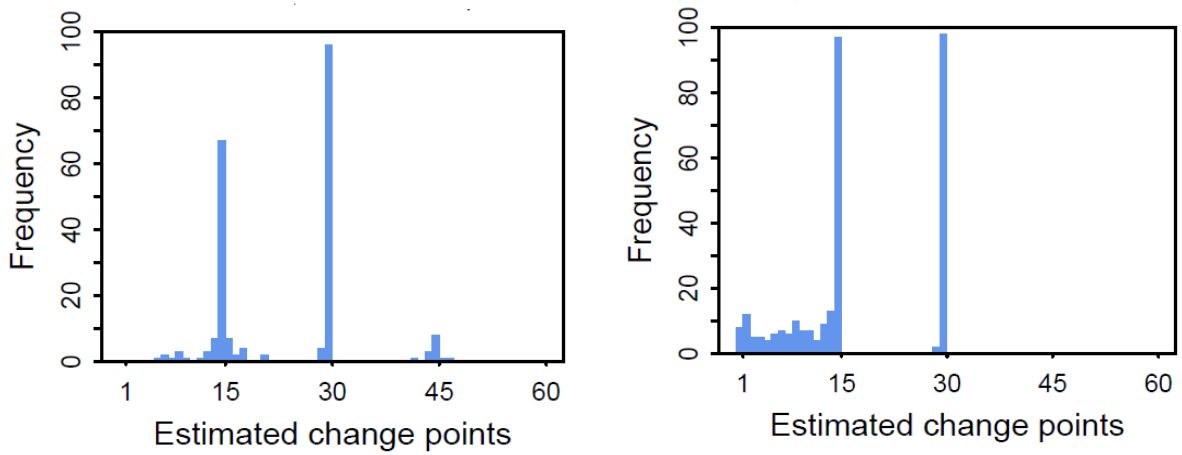


Fig. 4.3: E-div and Kernel-based method in Example 12

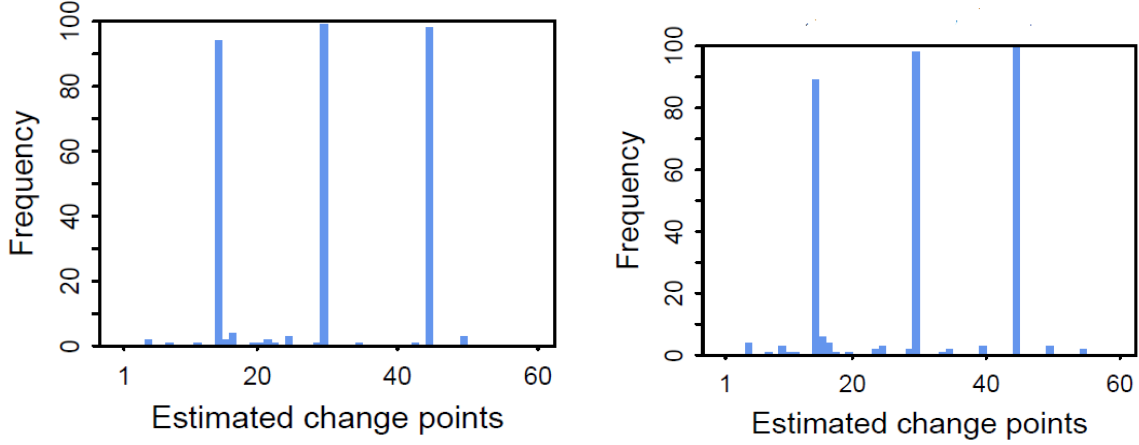


Fig. 4.4: DUNN index and Penalized DUNN in Example 13

Result of real-life data We apply our proposed method to the Reality Mining data set gathered during a research study conducted by the MIT Media Laboratory, encompassing a diverse group of individuals, including university students and staff members. The data collection took place between July 20, 2004, and June 14, 2005, utilizing mobile phones equipped with pre-installed software that logged call details. The complete dataset can be accessed at the MIT Media Laboratory in this. Website(<http://realitycommons.media.mit.edu/realitymining.html>). It comprises comprehensive information regarding the caller, callee, and timestamp for each call made throughout this specific timeframe. The primary inquiry revolves around identifying any potential alterations in the phone call patterns within this dataset. In our analysis, information on Candidate 1,2,3 is missing. we first eliminate observations containing missing values and focus on the available data pertaining to 84 individuals. Using this refined data set, we create an $(84, 84)$ dimensional matrix for each of the 48 weeks. Within these matrices, the entry at position (i, j) is assigned a value of the number of phone calls between the i -th and j -th individuals during that specific week. To further our analysis, we consider the $\binom{84}{2} = 3486$ distinct elements of the adjacency matrix, derived from the 84 individuals, as components of a 3486-dimensional vector.

Our proposed methodologies, DI and PDI both identified week 22 (December 14-20, 2004) as the change point. In contrast, the E-divisive method recognized week 7 (August 31 - September 06, 2004) as the change point, while the kernel method detected week 36 (March 22-28, 2005) as the pivotal period. the graph-based techniques, including MST, NNG, and MDP, the change-point was detected at week 41 (April 26 - May 02, 2005), week 33 (February 28 - March 07, 2005), and week 4 (August 10-16, 2004), correspondingly. It is important to note that the 22nd week marked the start of the winter break, during which

students and staff transitioned from the fall session to the spring session. This contextual information provides compelling justification for selecting week 22 as the change point in the friendship network analysis.

5. CONCLUSION

In conclusion, the change point data analysis method utilizing clustering has proven to be a valuable tool for detecting and understanding shifts in data patterns. Through the application of clustering techniques, we were able to identify significant points of change within the dataset, allowing us to gain insights into underlying dynamics and transitions.

By employing clustering algorithms, such as k-means or hierarchical clustering, we were able to group data points with similar characteristics and identify distinct clusters representing different periods or regimes. This enabled us to effectively detect and analyze change points where shifts in data behaviour occurred. The clustering-based approach provided several advantages in change point analysis. It allowed for unsupervised identification of change points without relying on predefined thresholds or assumptions. Furthermore, it offered flexibility in accommodating different types of data and patterns, making it applicable across various domains and datasets.

The results obtained from the change point data analysis method using clustering have contributed to a deeper understanding of the dataset dynamics, uncovering meaningful transitions and providing valuable insights for decision-making and further analysis. This method has demonstrated its effectiveness in addressing change point detection challenges and holds promise for future applications in diverse fields such as finance, healthcare, and environmental monitoring.

Overall, the utilization of clustering-based methods for change point analysis has showcased its significance in uncovering hidden patterns, identifying shifts in data behaviour, and facilitating a more comprehensive understanding of complex datasets. Continued research and advancements in clustering algorithms and techniques will further enhance the accuracy and applicability of this approach, empowering researchers and analysts to extract meaningful information from data and drive informed decision-making.

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